

API Setup Progress and Evidence Log

Detailed implementation trail for the StackEval benchmark prototype

Project	StackEval Benchmark - Professional Research Project
Document type	Progress and Evidence Log
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Purpose of this document

This document records the practical step-by-step work completed while building the first working StackEval prototype. It complements the decision log by making the implementation trail, setbacks, and fixes visible as evidence.

Final outcome

- OpenAI worked successfully with GPT-5.4 and GPT-5.4-mini.
- Anthropic worked successfully with Claude Sonnet 4.6.
- OpenRouter worked successfully with Llama 4 Maverick.
- A combined script was created so one question can be sent to all four project models in one run.
- Outputs were stored in JSON and readable TXT files.

Step-by-step progress log

Step	Action	Issue faced	Fix / decision	Evidence produced
1	Created OpenAI API account and generated first API key.	No immediate technical issue, but billing was not yet configured.	Stored the key in a .env file as the first secure setup step.	OpenAI key; .env file created.
2	Wrote the first OpenAI Python test script.	openai package was not found in the active Python environment.	Installed packages with the same Python interpreter used to run the script.	Successful package installation.
3	Ran the script again.	Environment variable was not read correctly because the key name did not match the code.	Changed the variable name to OPENAI_API_KEY in the .env file.	Correct environment configuration.
4	Tried the first real OpenAI call.	Received insufficient_quota / 429 because API billing was not active.	Recognized that ChatGPT access is separate from API billing.	Operational understanding of provider billing.
5	Tried to activate billing.	No credit card was available.	Created a Revolut account and used a virtual Visa card to	Billing enabled; OpenAI call succeeded.

Step	Action	Issue faced	Fix / decision	Evidence produced
			add API credits.	
6	Received the first successful OpenAI answer.	Terminal output alone was not enough for later evaluation.	Added JSON storage and human-readable TXT output.	openai_test_result.json and openai_readable.txt.
7	Created Anthropic API key and test script.	Needed a different SDK and provider-specific call format.	Installed anthropic package and adapted the script structure.	Working Claude script.
8	Verified Claude model choice.	There was uncertainty about Sonnet 4.5 versus 4.6 naming and availability.	Checked Anthropic release/docs and aligned the project choice to Sonnet 4.6.	Working Anthropic output.
9	Created OpenRouter API key and test script for the open model.	Needed a practical API route for Meta Llama.	Used OpenRouter as the access layer and selected Llama 4 Maverick as the open-model representative.	Working OpenRouter output.
10	Reviewed model fit against the project categories.	Needed a final consistent set of four models.	Confirmed the benchmark set: GPT-5.4, Claude Sonnet 4.6, Llama 4 Maverick, GPT-5.4-mini.	Final selected model set.
11	Built the combined multi-model script.	Previously each provider ran in a separate file.	Merged provider logic into one script that asks one question and returns four outputs in a row.	Combined benchmark prototype script.
12	Prepared outputs for later human scoring.	Raw JSON is not comfortable for developers to read directly.	Kept JSON for structure and TXT for readability.	Two output formats ready for evaluation.

Problems that became strong evidence

- Module installation error: proves that the setup was tested in a live environment instead of only planned on paper.
- Environment variable mismatch: proves that secure key handling was actually implemented and debugged.
- Insufficient quota: proves that the API request path was correct before billing was solved.
- No credit card available: shows real operational constraints and how they were solved pragmatically.
- Need for readable outputs: shows attention to the needs of later human evaluators, not just technical completion.

Websites and platforms used during the process

Platforms, docs, and references

Source	Why it was used	Link
OpenAI Platform	Created key, activated billing, and tested the first API-based model.	https://platform.openai.com/
OpenAI developer quickstart	Used as setup guidance for the first API request.	https://developers.openai.com/api/docs/quickstart/
Anthropic Console	Created Claude API key and tested the Anthropic route.	https://console.anthropic.com/
Anthropic Sonnet 4.6 release	Used to justify the latest Sonnet model choice.	https://www.anthropic.com/news/claude-sonnet-4-6
OpenRouter	Created key and accessed Llama 4 Maverick through API.	https://openrouter.ai/
OpenRouter Llama 4 Maverick page	Used to select the open-model route for the project.	https://openrouter.ai/meta-llama/llama-4-maverick
Meta Llama 4 announcement	Used to understand the difference between Scout and Maverick.	https://ai.meta.com/blog/llama-4-multimodal-intelligence/
Prosus StackEval GitHub	Used to align the project with a real benchmark	https://github.com/ProsusAI/stack-eval

Source	Why it was used	Link
	approach and reproducibility mindset.	
StackEval paper	Used as the academic reference for the benchmark concept.	https://openreview.net/forum?id=42mqplrA39
Revolut	Used operationally to create a virtual card and solve billing access.	https://www.revolut.com/

How this supports the portfolio

- It creates visible implementation evidence rather than only describing intentions.
- It supports LO1 by showing API feasibility research and provider comparison.
- It supports LO3 by showing design choices around reproducibility and output structure.
- It supports LO4 by showing working code and successful API connections.
- It supports LO5 in an early form by showing monitoring, debugging, and iterative improvement.